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%% =====
% Name      : Suraj Kumar Jha
% Roll No.  : 231537
% Experiment : Basic Discrete-Time Signals
% Description : Plotting various discrete-time signals
%% =====

clc;
clear;
close all;

%% =====
% Signal Generation
%% =====

n = -5:5;

% Unit Step Signal
unit_step = (n >= 0);

% Unit Impulse Signal
impulse = (n == 0);

% Ramp Signal
ramp = n .* unit_step;

% Signum Signal
sgn_signal = sign(n);

% Exponential Increasing Signal
exp_inc = 4*exp(0.2*n);

% Exponential Decreasing Signal
exp_dec = 4*exp(-0.2*n);

%% =====
% Figure 1 : Unit Step, Impulse and Ramp
%% =====

figure;

subplot(3,1,1)
stem(n,unit_step,'filled','LineWidth',1.5)
grid on
xlabel('Time Index (n)')
ylabel('Amplitude')
title('Unit Step Signal')
ylim([-0.5 1.5])

subplot(3,1,2)

```

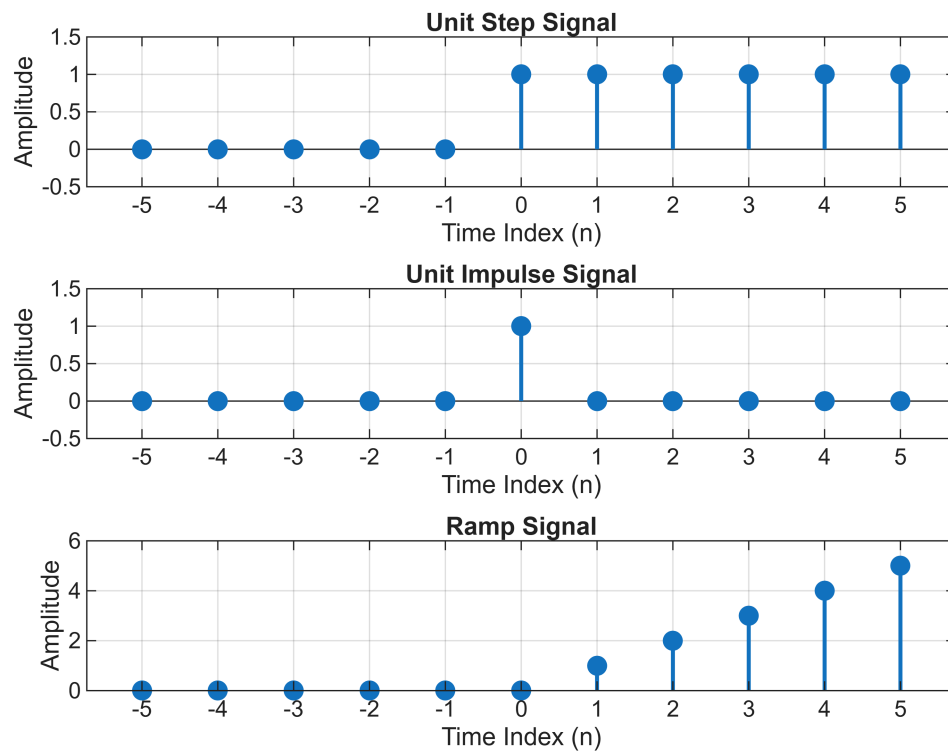
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stem(n,impulse,'filled','LineWidth',1.5)
grid on
xlabel('Time Index (n)')
ylabel('Amplitude')
title('Unit Impulse Signal')
ylim([-0.5 1.5])

subplot(3,1,3)
stem(n,ramp,'filled','LineWidth',1.5)
grid on
xlabel('Time Index (n)')
ylabel('Amplitude')
title('Ramp Signal')

exportgraphics(gcf,'Figure1_BasicSignals.png','Resolution',300);

```



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%% =====
% Figure 2 : Signum and Exponential Signals
%% =====

figure;

subplot(3,1,1)
stem(n,sgn_signal,'filled','LineWidth',1.5)
grid on
xlabel('Time Index (n)')

```

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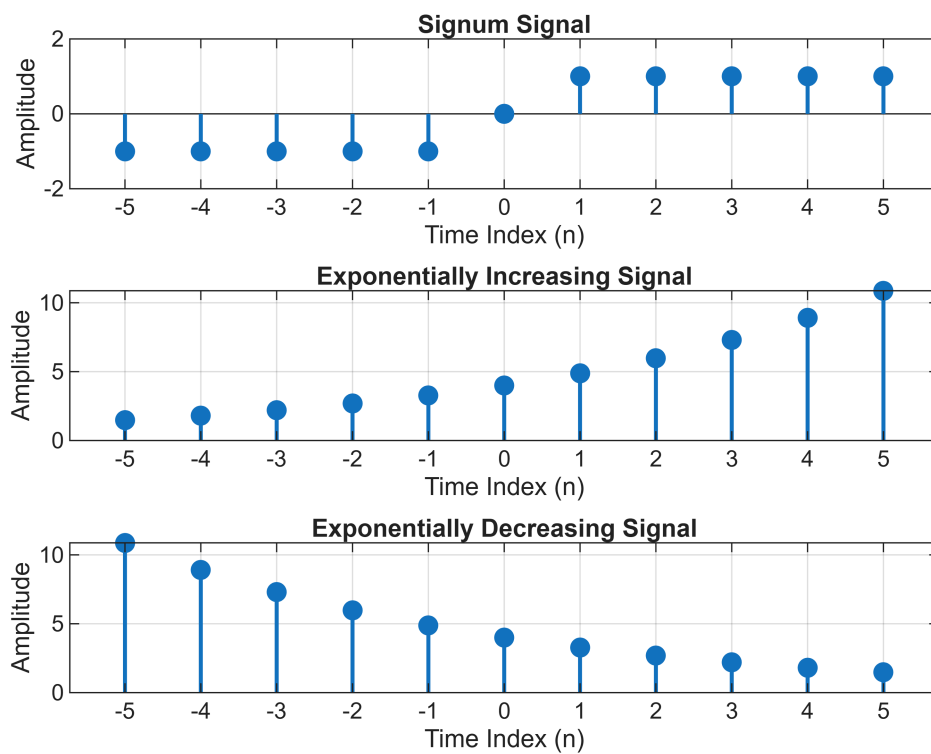
ylabel('Amplitude')
title('Signum Signal')
ylim([-2 2])

subplot(3,1,2)
stem(n,exp_inc,'filled','LineWidth',1.5)
grid on
xlabel('Time Index (n)')
ylabel('Amplitude')
title('Exponentially Increasing Signal')

subplot(3,1,3)
stem(n,exp_dec,'filled','LineWidth',1.5)
grid on
xlabel('Time Index (n)')
ylabel('Amplitude')
title('Exponentially Decreasing Signal')

exportgraphics(gcf,'Figure2_ExponentialSignals.png','Resolution',300);

```



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%% =====
% End of Program
%% =====

disp('-----');

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disp('Basic Discrete-Time Signals Plotted');
```

Basic Discrete-Time Signals Plotted

```
disp('Figures exported successfully.');
```

Figures exported successfully.

```
disp('-----');
```